Roll No:

ODD END SEMESTER EXAMINATION DECEMBER – 2024

Name of the Course: M. TECH CAD/CAM

Semester: I

Name of the Paper: Advanced Machine Design

Paper Code: DEME11

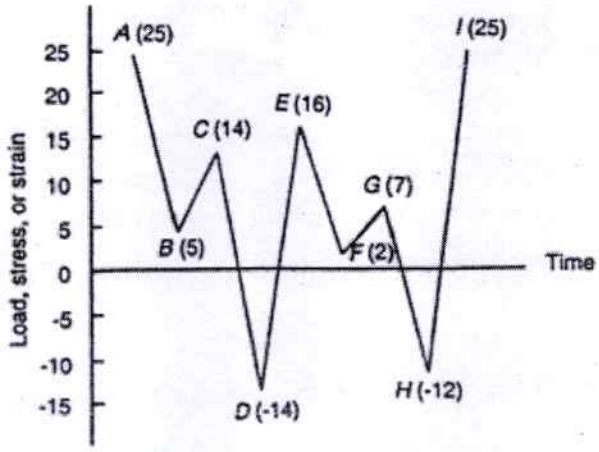
Time: 3 Hours

Maximum Marks: 100

Note: -

(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks

Q1		**(20marks)**
(a)	Explain different 'Modes of Failure' of a mechanical component.	CO1, CO3
(b)	Explain Strain hardening or work hardening with the help of Stress-strain Diagram?	
(c)	What do you mean by fatigue failure? Explain mechanism of fatigue fracture.	
Q2		**(20 marks)**
(a)	Explain behavior of crack under 'static loading' and 'fatigue loading'?	CO3
(b)	What do you mean by 'Fatigue life'	
(c)	Explain ASTM Experiment to obtain 'Crack growth vs No. of cycles' of notched specimen.	
Q3		**(20 marks)**
(a)	Fatigue failure starts from initiation of crack in the material. The crack initiates generally at discontinuities like change in cross-section, holes, notches, defects, impurities in the material etc. How you can explain the initiation of crack in ideal material (free from any cut, holes, defects, impurities)?	CO2,CO3
(b)	What is 'Fractography'? Explain and compare the general fractograph of ductile, brittle and fatigue fracture.	
(c)	Define the following in context of Fatigue of metals. (i) SN Diagram of steel (ii) Endurance Limit	

Q4		(20 marks)
(a)	What do you mean by cumulative damage? Derive and explain Miner's rule.	CO4, CO5
(b)	Explain briefly various fatigue life models.	
(c)	Explain the estimation of fatigue life using counting method.	
Q5		**(20 marks)**
(a)	Reduce the following spectrum of varying stress into a set of simple stress reversals using Rainflow counting method. 	CO5, CO6
(b)	Explain 'Paris law' for crack propagation. Write the 'Paris equation' for different materials and derive the expression of 'life'.	
(c)	Explain 'Stress intensity factor' and 'Fracture toughness'.	